Informatica® Cloud Data Integration

Amazon SageMaker Lakehouse Connector

Informatica Cloud Data Integration Amazon SageMaker Lakehouse Connector
October 2026

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Table of Contents

Preface 4

Informatica Resources 4

- Informatica Documentation 4
- IDMC web site 4
- IDMC Communities 4
- IDMC Marketplace 4
- Data Integration connector documentation 5
- Informatica Knowledge Base 5
- IDMC Trust Center 5
- IDMC Pulse Dashboard 5
- Informatica Global Customer Support 5

Chapter 1: Introduction to Amazon SageMaker Lakehouse Connector 6

Amazon SageMaker Lakehouse Connector assets 6

Chapter 2: Connections for Amazon SageMaker Lakehouse 7

Prerequisites 7

- Create minimal IAM policies 7
- Configure EC2 role to assume role 9
- Configure cross-account access for EC2 role to assume role authentication 10

Connect to Amazon SageMaker Lakehouse 15

- Before you begin 15
- Connection details 15
- Storage authentication type 16
- Advanced settings 17

Chapter 3: Mappings for Amazon SageMaker Lakehouse 18

Prerequisites for mappings 18

Sources for Amazon SageMaker Lakehouse 19

Targets for Amazon SageMaker Lakehouse 20

Chapter 4: Data type reference 23

Amazon SageMaker Lakehouse and transformation data types 23

Index 25

Preface

Use *Amazon SageMaker Lakehouse Connector* to learn how to read from or write to Apache Iceberg tables by using Cloud Data Integration. Learn to create an Amazon SageMaker Lakehouse connection, develop and run mappings in advanced mode, and mapping tasks in Cloud Data Integration.

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CHAPTER 1

Introduction to Amazon SageMaker Lakehouse Connector

You can use Amazon SageMaker Lakehouse Connector to read from and write to Apache Iceberg tables, that are managed by the AWS Glue Data Catalog or S3 tables catalog and stored in Amazon S3, within Amazon SageMaker Lakehouse.

You can use an Amazon SageMaker Lakehouse connection as a source or target only in mappings in advanced mode.

When you configure an Amazon SageMaker Lakehouse connection, you can select the lakehouse pattern that is a combination of catalog type and storage type that you want to connect to. To read from an AWS Glue Data catalog, select the S3 Data Lake lakehouse pattern. To read from an S3 tables catalog, select the S3 Tables lakehouse pattern.

You can configure cross-account access when the Secure Agent running on an EC2 instance is in one AWS account, while Apache Iceberg tables, Glue Catalogs, and Athena workgroups reside in a different AWS account.

Amazon SageMaker Lakehouse Connector assets

Create assets in Data Integration to integrate data using Amazon SageMaker Lakehouse Connector.

When you use Amazon SageMaker Lakehouse Connector, you can include the following Data Integration assets:

- Mapping in advanced mode
- Mapping task

CHAPTER 2

Connections for Amazon SageMaker Lakehouse

Create an Amazon SageMaker Lakehouse connection to securely read from and write to Apache Iceberg tables that are managed by the AWS Glue Data catalog or S3 tables catalog and stored in Amazon S3.

Before you create a connection, enable the connector for the runtime environment. For more information about enabling connectors, see *Service and connector assignment* in the Administrator help.

You can use an Amazon SageMaker Lakehouse connection to specify sources and targets in mappings in advanced mode and mapping tasks.

Prerequisites

Before you create an Amazon SageMaker Lakehouse connection, you need to create IAM policies with the minimum required permissions to interact with Apache Iceberg tables managed by AWS Glue Catalog or S3 tables catalog.

To use EC2 role to assume role authentication for Amazon S3, you need to configure the EC2 role to assume another IAM role specified by the IAM Role ARN.

Create minimal IAM policies

You need to create IAM policies with the minimum required permissions to interact with Apache Iceberg tables managed by AWS Glue Catalog or S3 tables catalog. For more information on configuring these policies, refer to the AWS documentation.

Minimum policy for Amazon Athena

The following sample policy shows the minimal Amazon IAM policy to access Amazon Athena:

```
{
    "Version": "2025-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "athena:CreatePreparedStatement",
                "athena:GetPreparedStatement",
                "athena:GetWorkGroup",
                "athena:GetTableMetadata",
                "athena:StartQueryExecution",
                "athena:GetQueryResultsStream",
```

```
                "athena:ListDatabases",
                "athena:GetQueryExecution",
                "athena:GetQueryResults",
                "athena:GetDatabase",
                "athena:ListTableMetadata",
                "athena:GetDataCatalog",
                "athena:DeletePreparedStatement"
            ],
            "Resource": [
                "arn:aws:athena:*:*:workgroup/*",
                "arn:aws:athena:*:*:datacatalog/*"
            ]
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "athena:ListDataCatalogs",
                "athena:GetQueryExecution",
                "athena:ListWorkGroups",
                "athena:GetPreparedStatement"
            ],
            "Resource": "*"
        }
    ]
}
```

Minimum policy for AWS Glue

The following sample policy shows the minimal Amazon IAM policy to access AWS Glue Catalog:

```
{
    "Version": "2025-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "glue:*"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Minimum policy for Amazon S3

The following sample policy shows the minimal Amazon IAM policy to read from or write data to an Amazon S3 bucket:

```
{
    "Version": "2025-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:ListBucket",
                "s3:DeleteObject"
            ],
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:ListAllMyBuckets",
```

```
                "s3:GetBucketAcl"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Minimum policy for S3 tables

The following sample policy shows the minimal Amazon IAM policy to read from or write data to S3 tables:

```
{
    "Version": "2025-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "sts:AssumeRole",
            "Resource": "arn:aws:iam::001234567890:role/S3TableSparkrole"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3tables:CreateTable",
                "s3tables:GetTable",
                "s3tables:ListTables",
                "s3tables:DeleteTable",
                "s3tables:GetTableMetadataLocation",
                "s3tables:GetTableData",
                "s3tables:PutTableData",
                "s3tables:UpdateTableMetadataLocation"
            ],
            "Resource": [
                "arn:aws:s3tables:us-east-1:001234567890:bucket/sagemaker-s3tables",
                "arn:aws:s3tables:us-east-1:001234567890:bucket/sagemaker-
s3tables/*",
                "arn:aws:s3tables:us-east-1:001234567890:bucket/sagemaker-s3tables/
table/*"
            ]
        }
    ]
}
```

Minimum policy for AWS Lake Formation

The following sample policy shows the minimal Amazon IAM policy to access AWS Lake Formation:

```
{
    "Version": "2025-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": "lakeformation:*",
            "Resource": "*"
        }
    ]
}
```

Configure EC2 role to assume role

You can configure an EC2 role to assume an IAM role and generate temporary security credentials to connect to Amazon S3 from the same or different AWS accounts.

When you configure an EC2 role to assume an IAM role, ensure that you have the **sts:AssumeRole** permission and a trust relationship established within the AWS accounts to use temporary security credentials. The trust relationship is defined in the trust policy of the IAM role when you create the role. The IAM role adds the EC2

role as a trusted entity allowing the EC2 role to use the temporary security credentials and access the AWS accounts.

When the trusted EC2 role requests for the temporary security credentials, the AWS Security Token Service (AWS STS) dynamically generates the temporary security credentials that are valid for a specified period and provides the credentials to the trusted EC2 role.

Before you use the EC2 Role to Assume Role authentication, consider the following prerequisites:

- Install the Secure Agent on the AWS EC2 instance.
- The EC2 role attached to the AWS EC2 instance must have permissions to assume another IAM role. The following code snippet a sample permission policy of EC2 role that is attached to the AWS EC2 instance:

```
{
    "Version": "2025-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "sts:AssumeRole",
            "Resource": "arn:aws:iam::001234567890:role/sagemaker_rolearn"
        }
    ]
}
```

  Resource value must include the ARN of IAM role that the EC2 role needs to assume.

- The IAM role that the EC2 role needs to assume must have a permission policy and a trust policy attached to access AWS Glue Catalog, Amazon Athena, and Amazon S3.
  You can also specify the external ID of your AWS account for a more secure access. The external ID must be a string.

  The following code snippet shows the assumed IAM role's trust policy with the external ID:

```
{
    "Version": "2025-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::001234567890:root" //anyone in the account
001234567890 can assume this role, this can also be limited to one role.
            },
            "Action": "sts:AssumeRole",
            "Condition": {
                "StringEquals": {
                    "sts:ExternalId": "aws_externalid"
                }
            }
        }
    ]
}
```

Configure cross-account access for EC2 role to assume role authentication

You can configure cross-account access when the Secure Agent running on an EC2 instance is in one AWS account, while Apache Iceberg tables, AWS Glue Catalogs, and Athena workgroups reside in a different AWS account.
Consider the following example to set up cross-account IAM roles and policies to allow an EC2 agent running in Account B to access AWS Glue Catalog, Athena workgroup, and Apache Iceberg tables located in Account A:

- Account A hosts the IAM Role to be assumed, AWS Glue Catalog, Athena workgroup, and Iceberg tables.

- Account B hosts the EC2 instance running the Secure Agent that needs to assume the role in Account A.

Perform the following steps to configure cross-account access for an EC2 instance:

1. Create IAM policies in Account A for AWS Glue Catalog, Amazon Athena, Amazon S3, and AWS Lake Formation.
2. Create an IAM role in Account A with trust policy allowing Account B to assume the role using an External ID.
3. Create an IAM policy in Account B allowing EC2 instance to assume the role in Account A.
4. Create and assign the IAM Role to the EC2 instance in Account B.
5. Grant AWS Lake Formation data permissions for Amazon SageMaker Lakehouse Connector.

Create IAM policies in Account A

You need to create IAM policies with the minimum required permissions to interact with AWS Glue Catalog, Amazon Athena, Amazon S3, and AWS Lake Formation.

1. Log in to the IAM console.
2. Click **Policies** > **Create policy**.
3. On the **JSON** tab, add the following policies:
   Minimum policy for Amazon Athena

   The following sample policy shows the minimal Amazon IAM policy to access Amazon Athena:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "athena:CreatePreparedStatement",
                "athena:GetPreparedStatement",
                "athena:GetWorkGroup",
                "athena:GetTableMetadata",
                "athena:StartQueryExecution",
                "athena:GetQueryResultsStream",
                "athena:ListDatabases",
                "athena:GetQueryExecution",
                "athena:GetQueryResults",
                "athena:GetDatabase",
                "athena:ListTableMetadata",
                "athena:GetDataCatalog",
                "athena:DeletePreparedStatement",
                "athena:ListWorkGroups",
                "athena:ListDataCatalogs"
            ],
            "Resource": [
                "arn:aws:athena:*:*:workgroup/*",
                "arn:aws:athena:*:*:datacatalog/*"
            ]
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "athena:ListDataCatalogs",
                "athena:GetQueryExecution",
                "athena:ListWorkGroups",
                "athena:GetPreparedStatement"
            ],
            "Resource": "*"
```

```
            }
        ]
    }
```

Minimum policy for AWS Glue

The following sample policy shows the minimal Amazon IAM policy to access AWS Glue Catalog:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "glue:*"
            ],
            "Resource": "*"
        }
    ]
}
```

Minimum policy for Amazon S3

The following sample policy shows the minimal Amazon IAM policy to read from or write data to an Amazon S3 bucket:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:ListBucket",
                "s3:DeleteObject"
            ],
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:ListAllMyBuckets",
                "s3:GetBucketAcl"
            ],
            "Resource": [
                "*"
            ]
        },
        {
            "Effect": "Allow",
            "Action": [
                "lakeformation:GetDataAccess",
                "lakeformation:ListPermissions",
                "lakeformation:ListResources"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Minimum policy for AWS Lake Formation

The following sample policy shows the minimal Amazon IAM policy to access AWS Lake Formation:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "LakeFormationApiAccess",
            "Effect": "Allow",
            "Action": [
                "lakeformation:GetDataAccess",
                "lakeformation:ListPermissions",
                "lakeformation:ListResources",
                "lakeformation:GetDataLakeSettings",
                "lakeformation:GetEffectivePermissionsForPath"
            ],
            "Resource": "*"
        }
    ]
}
```

Create an IAM role in Account A

Create an IAM role in Account A with trust policy allowing Account B to assume the role using an External ID.

1. In the IAM console, click **IAM** > **Roles** > **Create Role**.
2. On the **Step 1: Select trusted entities** tab, under the **Trusted entity type** section, select **AWS account**.
3. In the **An AWS account** section, select **Another AWS account**, and enter the Account ID for Account B that hosts the EC2 instance where the Secure Agent is installed.
4. Under **Options**, select **Require External ID** and enter a secure external ID.
5. Click **Next**.
6. On the **Step 2 Add permissions** tab, select and attach the created permission policies to access AWS Glue Catalog, Amazon Athena, Amazon S3, and AWS Lake Formation.
7. Click **Next**.
8. On the **Step 3 Name, review, and create** tab, enter a **Name** for the role.
9. Click **Create role**.
   After you create a role, copy the IAM role ARN. You must specify the IAM role ARN in the Amazon SageMaker Lakehouse connection to use cross-account access for EC2 Role to Assume Role authentication.
10. On the **Trust relationships** tab, you can view trust relationship policy for the IAM role.
    The following example shows the trust relationship policy for the IAM role:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": [
                    "arn:aws:iam::123456789101:root"
                ]
            },
            "Action": "sts:AssumeRole",
            "Condition": {
                "StringEquals": {
                    "sts:ExternalId": "sagemaker"
                }
            }
        }
```

```
        ]
    }
```

Create an IAM policy in Account B

Create an IAM policy in Account B allowing the EC2 instance to assume the IAM role in Account A.

1. In the IAM console, click **Policies** > **Create policy**.
2. On the **JSON** tab, add the following policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": "sts:AssumeRole",
            "Resource": "arn:aws:iam::AccountAID:role/assumerolename"
        }
    ]
}
```

   The value of the **Resource** field in the policy is the ARN of the IAM Role created in Account A. Replace `arn:aws:iam::AccountAID:role/assumerolename` with the actual IAM role ARN.

3. Click **Next**.
4. Name the policy, and then click **Create Policy**.

Create and assign the IAM Role to the EC2 instance in Account B

To create an IAM role for the EC2 instance in Account B, perform the following steps:

1. In the IAM console, click **IAM** > **Roles** > **Create Role**.
2. On the **Step 1: Select trusted entities** tab, under the **Trusted entity type** section, select **AWS service**.
3. Under **Use case**, select **EC2**.
4. Click **Next**.
5. On the **Step 2 Add permissions** tab, select and attach the created policy in Account B.
6. Click **Next**.
7. On the **Step 3 Name, review, and create** tab, enter the **Name** and **Description** for the role.
8. Click **Create role**.

To assign the IAM Role to the EC2 instance, perform the following steps:

1. Log in to the EC2 Console.
2. Select your instance, and then click **Actions** > **Security** > **Modify IAM role**.
3. Select the role you created, and then click **Update IAM role**.

Optionally, to whitelist an IP address, perform the following steps:

1. On the **Security** tab, click on the ID of the attached **Security Group**.
2. Click **Edit inbound rules**.
3. Click **Add rule**, select the **Protocol** and **Port range**.
4. Under **Source**, select **Custom** and enter the specific IP address you want to whitelist.
5. Click **Save rules**.

Grant AWS Lake Formation data permissions

Grant Lake Formation data permissions for Amazon SageMaker Lakehouse connector.

1. Open the AWS Lake Formation console in the region where SageMaker Studio is set up.
2. Click **Data permissions** > **Grant**.
3. Under **Principals**, select **IAM users and roles**.
4. In the **IAM users and roles** field, select the IAM role that you created in Account A.
5. Under **LF-Tags or catalog resources**, select **Named data catalog resources**.
6. Select the **Catalog** and **Database** where the Iceberg tables are stored.
7. Under **Database permissions**, select the following permissions:
   - Create table
   - Alter
8. Click **Grant**.

Connect to Amazon SageMaker Lakehouse

Let's configure the Amazon SageMaker Lakehouse connection properties to connect to Amazon SageMaker Lakehouse and read Apache Iceberg tables managed by AWS Glue Catalog or S3 tables catalog.

Before you begin

Before you get started, you need to create IAM policies with the minimum required permissions to interact with Apache Iceberg tables and configure the authentication-specific prerequisites for Amazon S3 storage.

Check out "Prerequisites" on page 7 to learn more about how to create policies and configure the authentication-specific prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent. You cannot use Hosted Agent, elastic runtime environment, and serverless runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Lakehouse Pattern	The pattern of Amazon SageMaker Lakehouse. Pattern is a combination of catalog type and storage type that you want to connect to. Select one of the following options: - S3 Data Lake. To read from and write to Apache Iceberg tables that are managed by the AWS Glue Data catalog and stored in Amazon S3. - S3 Tables. To read from and write to Apache Iceberg tables that are managed by the S3 tables catalog and stored in Amazon S3.

Athena JDBC URL	The JDBC URL to connect to Amazon Athena.
	For S3 Data Lake lakehouse pattern, enter the JDBC URL in the following format:
	`jdbc:athena://Region=<AWS_Region>;OutputLocation=<S3_Location>`
	For S3 Tables lakehouse pattern, enter the JDBC URL in the following format:
	`jdbc:athena://AwsRegion=us-east-1;Catalog=s3tablescatalog/your-bucket-name;Schema=your_namespace;`
	Where `s3tablescatalog/your-bucket-name` is your S3 table bucket catalog and `your_namespace` is the namespace where your table is stored.

Storage authentication type

You can use Permanent IAM Credentials or EC2 Role to Assume Role authentication types to connect to Amazon S3 storage.

Permanent IAM Credentials authentication

The Permanent IAM Credentials authentication requires the access key and secret key values to connect to Amazon S3 storage.

The following table describes the properties to configure the Permanent IAM Credentials authentication:

Property	Description
Access Key	The AWS Access Key ID that uniquely identifies your IAM user credentials to access Amazon S3 storage.
Secret Key	The AWS Secret Access Key that authenticates your Access Key ID for secure access to your Amazon S3 data.

EC2 Role to Assume Role authentication

The EC2 Role to Assume Role authentication requires the ARN of the IAM role the EC2 role assumes.

The following table describes the property to configure the EC2 Role to Assume Role authentication:

Property	Description
IAM Role ARN	The ARN of the IAM role that the EC2 role assumes to generate the temporary security credentials.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
External ID	A unique, user-defined string value that the IAM role requires the EC2 role to provide when calling the sts:AssumeRole API.

CHAPTER 3

Mappings for Amazon SageMaker Lakehouse

Use the Data Integration Mapping Designer to define and configure a mapping in advanced mode for Amazon SageMaker Lakehouse sources and targets.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

When you run a mapping in advanced mode, Data Integration analyzes the data logic defined in the mapping to automatically assign data logic to run on an advanced cluster.

Prerequisites for mappings

Before you configure and run a mapping, complete the required prerequisites.

When you read from and write to Apache Iceberg tables, you can specify the Iceberg Spark properties to configure the Spark behavior for the Iceberg tables at runtime.

When the source and target are located in different regions, you must specify the Amazon S3 bucket ARN property in the Iceberg Spark properties in the Advanced source and target properties.

Create ARN for Amazon S3 bucket

To create the Amazon S3 bucket ARN, perform the following steps:

1. Sign in to the AWS Management Console and open the Amazon S3 console.
2. In the navigation bar, select the AWS region where you want to create an access point for the Amazon S3 bucket.
3. In the left navigation pane, select **Access Points**.
4. Click **Create access point**.
5. Enter a unique name for the access point.
6. Select the Amazon S3 bucket that the access point is associated with.
7. In the **Network origin** field, select **Internet**.
8. Click **Create access point**.
9. After the access point is created, click on the name to view the properties.
10. Copy the Amazon Resource Name (ARN) property value.

Sources for Amazon SageMaker Lakehouse

When you configure a mapping in advanced mode to use an Amazon SageMaker Lakehouse source, you can configure the source properties.

Specify the name and description of the Amazon SageMaker Lakehouse source. Configure the source and advanced source properties for the Amazon SageMaker Lakehouse object.

The following table describes the Amazon SageMaker Lakehouse source properties that you can configure in a Source transformation:

Property	Description
Connection	Name of the source connection. You can select an existing connection, create a new connection, or define parameter values for the source connection property. If you want to overwrite the source connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Source Type	Type of the Amazon SageMaker Lakehouse source object. You can choose from the following source types: - Single Object. Select to specify a single Amazon SageMaker Lakehouse object. - Parameter. Select to specify a parameter name. You can configure the source object in a mapping task associated with a mapping that uses this Source transformation.
Parameter	A parameter file where you define values that you want to update without the need to edit the task. Select an existing parameter for the source object or click **New Parameter** to define a new parameter for the source object. The **Parameter** property appears only if you select parameter as the source type. If you want to overwrite the parameter at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.
Object	Source object for the mapping.

The following table describes the Amazon SageMaker Lakehouse query options that you can configure in a Source transformation:

Property	Description
Filter	Filters records based on the filter condition. Click **Configure** to add conditions to filter records and reduce the number of rows that the Secure Agent reads from the source. You can specify the following filter conditions: - Not Parameterized. Use a basic filter to specify the object, field, operator, and value to select specific records. - Completely Parameterized. Use a parameter to represent the field mapping. - Advanced. Use an advanced filter to define a more complex filter condition that uses the Amazon SageMaker Lakehouse query format. A mapping fails at run time when you specify a simple filter with `Starts With`, `Ends With`, or `Contains` operators to retrieve data. To avoid this issue, configure an advanced filter with a `like` query to filter query results based on patterns instead of exact textual matches.
Sort	Not applicable.

The following table describes the Amazon SageMaker Lakehouse source advanced properties that you can configure in a Source transformation:

Property	Description

Iceberg Spark Properties	The Spark configuration properties as key-value pairs that you want to configure for the Iceberg tables at runtime.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use the S3 Tables lakehouse pattern, you must specify the S3 table bucket ARN property in the following format:
	`TableBucketARN=arn:aws:s3tables:us-east-1:001234567890:bucket/sagemaker-s3tables`
	When the source and target are in different regions, you must specify the Amazon S3 bucket ARN property in the following format:
	`s3.access-points.<bucket name>=<S3 bucket ARN>`
	For more information about how to create the ARN for S3 bucket, see "Prerequisites for mappings" on page 18.

Pre-SQL	The SQL queries to run before reading data from Apache Iceberg tables. Ensure that the SQL queries use a valid Spark SQL syntax. You can enter multiple queries separated by a semicolon. You must prefix the table name with the `<LakehouseCatalog>` string and the database name. For example, `<LakehouseCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the pre-SQL query fails, the mapping also fails.
Post-SQL	The SQL queries to run after reading data from Apache Iceberg tables. Ensure that the SQL queries use a valid Spark SQL syntax. You can enter multiple queries separated by a semicolon. You must prefix the table name with the `<LakehouseCatalog>` string and the database name. For example, `<LakehouseCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the post-SQL query fails, the mapping also fails. If the mapping fails, the post-SQL query is not executed.
Tracing Level	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization, or verbose data. Default is normal.

Targets for Amazon SageMaker Lakehouse

When you configure a mapping in advanced mode to use an Amazon SageMaker Lakehouse target, configure the target properties.

Specify the name and description of the Amazon SageMaker Lakehouse target. Configure the target and advanced target properties for the Amazon SageMaker Lakehouse object in a Target transformation.

The following table describes the properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection. You can select an existing connection, create a new connection, or define parameter values for the target connection property. If you want to overwrite the target connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Target Type	Type of the Amazon SageMaker Lakehouse target object. You can choose from the following target types: - Single Object. Select to specify a single Amazon SageMaker Lakehouse object. - Parameter. Select to specify a parameter name. You can configure the target object in a mapping task associated with a mapping that uses this Target transformation.
Parameter	A parameter file where you define values that you want to update without having to edit the task. Select an existing parameter for the target object or click **New Parameter** to define a new parameter for the target object. The **Parameter** property appears only if you select parameter as the target type. If you want to overwrite the target object at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.
Object	Name of the target object. You can select an existing object from the list or create a target object at run time.
Operation	Type of the target operation. Select one of the following operations: - Insert - Update - Upsert - Delete You cannot use update, upsert, and delete target operations for S3 Tables lakehouse pattern. **Note:** You cannot configure the Data Driven operation on an Amazon SageMaker Lakehouse target.
Update Columns	The primary key columns to update, upsert, or delete data in a Amazon SageMaker Lakehouse target.

The following table describes the advanced properties that you can configure in a Target transformation:

Property	Description
Truncate Target	Truncates the target table before loading the data. By default, the property is not selected.

Iceberg Spark Properties	The Spark configuration properties as key-value pairs that you want to configure for the Iceberg tables at runtime.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use the S3 Tables lakehouse pattern, you must specify the S3 table bucket ARN property in the following format:
	`TableBucketARN=arn:aws:s3tables:us-east-1:001234567890:bucket/sagemaker-s3tables`
	When the source and target are in different regions, you must specify the bucket region property in the following format for the update, upsert, or delete target operations:
	`BucketRegion=<Amazon-S3-target-bucket-region-name>`
	For more information about how to create the ARN for S3 bucket, see "Prerequisites for mappings" on page 18.

Update Mode	Determines how the records are updated in the target table. This property applies when you select the **Update** or **Upsert** target operation. Select one of the following modes: - Update As Update. Updates records in the target table if the specified unique key column value matches with the incoming column value. - Update Else Insert. Updates records in the target table if the specified unique key column value matches with the incoming column value. If the unique key column value does not match, the mapping inserts a new row with the records.
Pre-SQL	The SQL queries to run before writing data to Apache Iceberg tables. Ensure that the SQL queries use a valid Spark SQL syntax. You can enter multiple queries separated by a semicolon. You must prefix the table name with the `<LakehouseCatalog>` string and the database name. For example, `<LakehouseCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the pre-SQL query fails, the mapping also fails.
Post-SQL	Post-SQL queries to run after writing data to Apache Iceberg tables. Ensure that the SQL queries use a valid Spark SQL syntax. You can enter multiple queries separated by a semicolon. You must prefix the table name with the `<LakehouseCatalog>` string and the database name. For example, `<LakehouseCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the post-SQL query fails, the mapping also fails. If the mapping fails, the post-SQL query is not executed.

CHAPTER 4

Data type reference

Data Integration uses the following data types in Amazon SageMaker Lakehouse mappings in advanced mode:

- **Amazon SageMaker Lakehouse native data types**: Amazon SageMaker Lakehouse data types appear in the source and target transformations when you choose to edit metadata for the fields.
- **Transformation data types**: Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mapping.
  When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Amazon SageMaker Lakehouse and transformation data types

The following table lists the Amazon SageMaker Lakehouse data types that Data Integration supports and the corresponding transformation data types:

Amazon SageMaker Lakehouse data type	Transformation data type	Description
Binary	Binary	Variable length binary data.
Boolean	String	True or false
Date	Date/Time	Calendar date without timezone and time. 0001-01-01 to 9999-12-31 Precision 29, scale 9
Decimal	Decimal	The maximum value for precision is 38, and the maximum value for scale is 38.
Double	Double	4.94065645841246544e-324d to 1.79769313486231570e+308d, positive or negative.

Amazon SageMaker Lakehouse data type	Transformation data type	Description
Float	Double	1.40129846432481707e-45 to 3.40282346638528860e+38, positive or negative.
Int	Integer	32-bit signed integers 2^{31} to 2^{31} -1.
Long	Integer	64-bit signed integers
String	String	Variable length character data.
Timestamp	Date/Time	Timestamp without timezone. Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision to the microsecond.

INDEX

C

Cloud Application Integration community
  URL 4
Cloud Developer community
  URL 4

D

Data Integration community
  URL 4

I

Informatica Global Customer Support
  contact information 5
Informatica Intelligent Data Management Cloud
  web site 4

M

maintenance outages 5

S

status
  IDMC 5
system status 5

T

trust site
  description 5

U

upgrade notifications 5

W

web site 4